Spot Micro - final parts list

31 Oct 2026. Replaces the 23 Oct list. Checked against the KDY0523 _cls frame in the 3D model. 3D-printed parts: see print/README.md.

1. Computing & sensors

	Item	Qty	Notes
☐	Raspberry Pi 3B	1	
☐	microSD card, 16-32 GB	1	
☐	Raspberry Pi Camera v2	1	
☐	Camera ribbon cable, 15-pin	1	30 cm (route is ~22 cm)
☐	MPU-6050 (IMU)	1	
☐	LCD 16x2 with I2C backpack	1	Address 0x27
☐	I2C level shifter, BSS138, 4-ch	1	LCD backpack pulls SDA/SCL toward 5 V; Pi GPIO is 3.3 V only
☐	Heatsink for Pi CPU	1	Skip if already installed
☐	Dupont jumper wires F-F	1 set	

2. Drive

	Item	Qty	Notes
☐	PCA9685 servo driver, 16-ch	1	
☐	Servo PDI-HV5523MG	12	
☐	Servo extension 3-pin, 20 cm	4	Knee servos
☐	Servo extension 3-pin, 10 cm	2	Right hip servos. Other 6 servos: own cable is enough

3. Power

	Item	Qty	Notes
☐	LiPo 2S 4000-5000 mAh, shorty hardcase	1	Must fit bay: max 127 x 48 x 25.5 mm (check height and width). Standard 138 mm packs do not fit. E.g. Gens Ace 2S 5000 mAh shorty, 95 x 46.8 x 25 mm
☐	UBEC 5 V / 5 A, 2S input	1	E.g. Hobbywing UBEC 5A (2-8S, 5 A cont.). Set output to 5.0 V before connecting the Pi
☐	Rocker switch Switch Components RF1-1A-DC-2-B	1	16 A at 12 V DC, panel cut-out 19.2 x 13 mm. 20 A rockers do not fit
☐	Blade fuse 15 A (ATC) + inline holder, 14 AWG	1	Not 20 A: switch is rated 16 A
☐	XT60 connector pair	1	Solder to the battery if it comes with bullets or T-plug
☐	XT60 Y-splitter	1	Servos + UBEC
☐	Capacitor 1000-2200 uF, 16 V or higher	1	On servo power line
☐	Silicone wire 14-16 AWG, red + black	1 m each	
☐	2S LiPo balance charger	1	
☐	LiPo low-voltage buzzer	1	

4. Voice

	Item	Qty	Notes
☐	USB microphone	1	
☐	USB speaker	1	

5. Optional: battery voltage

	Item	Qty	Notes
☐	ADS1115 ADC	1	
☐	Resistors 20 kOhm + 10 kOhm	1 + 1	Divider: 8.4 V -> 2.8 V

6. Fasteners & bearings (lengths measured on our _cls parts)

	Item	Qty	Notes
☐	Bolt M5x15	8	Covers to shoulder bulkheads
☐	Nut M5	8	Hex pockets in bulkheads (not in KDY BOM)
☐	Bolt M4x20	40	Servo mounting: shoulder, hip, knee
☐	Bolt M4x25	8	Knee servo, 20 mm holes in the _cls shin
☐	Nut M4	48	
☐	Bolt M3x25	24	Bulkheads (12); thigh + cover (12, check at assembly)
☐	Bolt M3x20	12	Side plates to bulkheads
☐	Nut M3	36	In nut slots
☐	Bolt M3x10 + nut M3	8 + 8	mike4192 shoulder reinforcement
☐	Bearing F625zz	12	Frame 8 + shoulder reinforcement 4
☐	Screw M2x6 + nut M2	4 + 4	Camera mount

7. Consumables

	Item	Qty	Notes
☐	Self-tapping screw 2 x 6-8 mm	8	Pi (4) and PCA9685 (4) to their printed platforms (holes 1.5 mm)
☐	Double-sided foam tape	1 roll	Pi and PCA9685 platforms to plate; MPU-6050 to plate
☐	Hot glue sticks + glue gun	1	Servo horns into printed parts (KDY0523 instructions)
☐	Zip ties, small	1 pack	Servo cables

Wiring

Same as the mike4192 build (2S direct to the PCA9685 servo terminal, UBEC 5 V for the Pi). Differences: PCA9685 logic at 3.3 V and a level shifter for the 5 V LCD, so the Pi I2C pins never see 5 V. The mike4192 software drives only the PCA9685 and the LCD; MPU-6050, camera, ADS1115 and voice need our own code.

Power (14-16 AWG)

From	To	Notes
LiPo 2S	XT60	Battery plug
XT60 (+)	Fuse 15 A -> switch -> Y-splitter (+)	Switch in the rear cover
XT60 (-)	Y-splitter (-)	
Y-splitter branch 1	PCA9685 V+ / GND terminal	Servo power 6.4-8.4 V; capacitor across this terminal
Y-splitter branch 2	UBEC input	
LiPo balance lead	Low-voltage buzzer	

5 V (UBEC output)

From	To	Notes
UBEC 5 V / GND	Raspberry Pi pin 2, 4 (5 V) / pin 6 (GND)	Output jumper at 5.0 V
UBEC 5 V / GND	LCD backpack VCC / GND, level shifter HV	

I2C bus, 3.3 V (Dupont)

From	To	Notes
Pi pin 1 (3.3 V)	PCA9685 VCC, MPU-6050 VCC, ADS1115 VDD, level shifter LV	PCA9685 logic at 3.3 V, not 5 V
Pi pin 3 SDA / pin 5 SCL	PCA9685 (0x40), MPU-6050 (0x68), ADS1115 (0x48)	No address conflicts
Pi pin 3 / 5 via level shifter	LCD SDA / SCL (0x27)	
Pi pin 9 (GND)	PCA9685 GND, all module GND	Common ground

Signals

From	To	Notes
PCA9685 channels	12 servos (3-pin plugs on the header)	Knee servos via 20 cm, right hips via 10 cm extensions
Battery (+) after switch	Divider 20k/10k -> ADS1115 A0	Optional, max 2.8 V at A0
Camera v2	Pi CSI connector, 30 cm ribbon	
USB mic, USB speaker	Pi USB ports	